SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1705**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I M. Mounika (192411083) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
M. Mounika

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

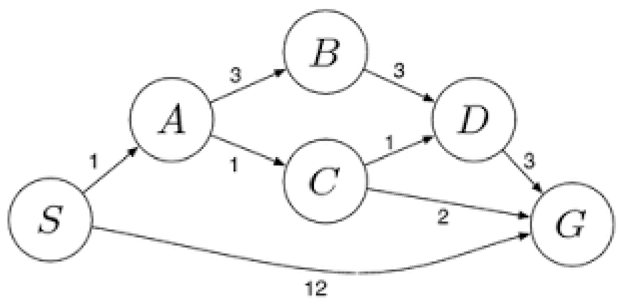
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
   i. What are the percept's for this agent?
   ii. Characterize the operating environment.
   iii. What are the actions the agent can take?
   iv. How can one evaluate the performance of the agent?
   v. What sort of agent architecture do you think is most suitable for this agent? Why?

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

   The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

SOLUTIONS

1. Water Jug Problem

Problem:

There are two jugs of 4 gallons and 3 gallons. The goal is to get exactly 2 gallons in the 4-gallon jug.

Assumptions:

- The jugs can be filled from the pump.
- Water can be emptied onto the ground.
- Water can be poured from one jug to another.
- No measuring device is available.

State Representation:

- (4-gallon jug, 3-gallon jug)

Solution Steps:

1. Start with both jugs empty.

State = (0,0)

2. Fill the 3-gallon jug.

State = (0,3)

3. Pour the 3-gallon jug into the 4-gallon jug.

State = (3,0)

4.Fill the 3-gallon jug again.

State = (3,3)

5.Pour water from the 3-gallon jug into the 4-gallon jug until it becomes full.

State = (4,2)

6.Empty the 4-gallon jug.

State = (0,2)

7.Pour the remaining 2 gallons from the 3-gallon jug into the 4-gallon jug.

State = (2,0)

Final Answer:

The 4-gallon jug contains exactly **2 gallons**.

2. Mars Rover Agent

i. What are the percepts?

- Camera images
- Rock samples
- Soil samples
- Temperature
- Atmospheric pressure
- Wind speed
- Terrain information
- Battery level
- Position and location

ii. Characterize the operating environment.

- It is partially observable.
- It is a single-agent environment.
- It is stochastic because the environment is uncertain.
- It is sequential because current actions affect future actions.
- It is dynamic because conditions change over time.
- It is continuous.
- It is mostly unknown.

iii. What actions can the agent perform?

- Move forward
- Move backward
- Turn left
- Turn right
- Stop
- Capture images

- Collect rock samples
- Analyze soil
- Send information to Earth
- Recharge using solar panels

iv. How is the performance evaluated?

- Number of samples collected
- Accuracy of scientific analysis
- Distance explored
- Battery efficiency
- Safe navigation
- Successful communication with Earth

v. Suitable agent architecture

The most suitable agent is a **Model-Based Goal-Based Agent**.

Reason:

- It stores information about the environment.
- It plans safe paths.
- It avoids obstacles.
- It works independently.
- It makes decisions to achieve its mission goals.

3. 8-Queens Problem

Problem Statement:

Place eight queens on an 8×8 chessboard so that no queen attacks another queen.

Problem Formulation

Initial State

- Empty chessboard.

State

- A board with some queens already placed.

Actions

- Place one queen in a safe position in the next column.

Transition Model

- After placing a queen, move to the next column.

Goal Test

- All 8 queens are placed.
- No two queens attack each other.

Path Cost

- Each move costs 1.
- Total cost = 8.

Search Strategy

- Depth First Search (DFS)
- Backtracking Algorithm

Steps

Step 1: Place the first queen.

Step 2: Move to the next column.

Step 3: Check whether the position is safe.

Step 4: If safe, place the queen.

Step 5: If not safe, try another row.

Step 6: If no row is safe, backtrack.

Step 7: Continue until all 8 queens are placed.

Final Answer:

The solution is obtained when all eight queens are placed without attacking each other.

4. OLA Cab Booking Application

Type of Problem-Solving Agent

The suitable agent is a **Goal-Based Agent**.

Reason

- The customer's goal is to reach the destination.
- The agent compares available cab options.
- It selects the best cab based on fare and comfort.

Pseudocode

START

Enter pickup location.

Enter destination.

Display available cab types.

Select cab type.

Calculate fare.

Display fare.

If customer confirms booking,

Assign the nearest driver.

Show driver details.

Start the trip.

Reach destination.

Generate bill.

Else,

Cancel booking.

End If.

STOP.

5. Uniform Cost Search (UCS)

Given Edge Costs

S → A = 1

S → G = 12

A → B = 3

A → C = 1

B → D = 3

C → D = 1

C → G = 2

D → G = 3

Start from the source node S.

The cost of S is 0.

Expand node S.

The possible paths are:

S → A = 1

S → G = 12

Choose A because it has the lowest cost (1).

Expand node A.

The new paths are:

S → A → B = 4

S → A → C = 2

Choose C because it has the lowest cost (2).

Expand node C.

The new paths are:

S → A → C → D = 3

S → A → C → G = 4

Choose D because it has the lowest cost (3).

Expand node D.

The new path is:

S → A → C → D → G = 6

The frontier now contains:

S → A → B = 4

S → A → C → G = 4

S → A → C → D → G = 6

S → G = 12

The first goal node reached with the minimum cost is:

S → A → C → G = 4

Optimal Path: S → A → C → G

Minimum Cost: 4

Final Answer

Optimal Path: S → A → C → G

Minimum Cost: 4